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1 Uncertainty, Entropy and Maximum Entropy Principle – An Overview

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Abstract. This introductory chapter aims to spell out the basics of entropy, maximum entropy frameworks and their applications in some fields. Applications are selected so as to avoid possible overlaps with the ones contained in this volume. Though the contents in this book are largely concerned with uncertainty of random phenomena, a brief overview of uncertainty manifesting in various other forms is also given.

1.1 Uncertainty

Uncertainty is ubiquitous and its understanding is central to decision making. Uncertainty emerges as one has less information than the total information required to describe a system and its environment. Uncertainty and information are so closely associated that the information provided by an experiment for example, is equal to the amount of uncertainty removed. It may be pertinent to point out that uncertainty manifests itself in several forms and various kinds of uncertainties may arise from random fluctuations, incomplete information, imprecise perception, vagueness etc. For instance, one encounters uncertainty due to vagueness in communication through natural language. Uncertainty in this sense is represented by fuzziness resulting from imprecision of meaning of a concept expressed by linguistic terms [1]. Fuzzy set concept provides an appropriate mathematical framework for dealing with vagueness. The framework based on probability theory has successfully been employed to deal with uncertainty of a random phenomenon. A commonly adopted approach to study stochastic evolution of a system is through master–equation formulation for the process in question. This approach takes into account intrinsic stochasticity which arises due to discreteness of variables of the system. An important aspect which is ignored in this type of modelling is that the environment in which the given system is embedded is endowed with a large number of degrees of freedom fluctuating randomly in time, giving rise to environmental stochasticity [2]. Probability theory only deals with uncertainty associated with the expectation of future event based on something new.

Since uncertainty represented by fuzziness is not the uncertainty of expectation, the probabilistic framework fails to capture it. The introduction

of evidence and possibility theories has enriched analytical tools and appear to be very promising for studying uncertainty on account of fuzziness and lack of specificity [3,4]. In the last few decades several mathematical models which are not based on probabilistic framework have been proposed [1,4]. The Chapter by Klir of this volume gives a good exposition of uncertainty that arises in an evidence theoretic framework. In this chapter we provide introductory materials relating to uncertainty associated with a probabilistic system. We also provide a brief account of uncertainties (fuzziness and non–specificity) associated with fuzzy systems. A brief introduction to information associated with a quantum mechanical system is also provided.

1.2 Measure of Uncertainty in Random Phenomena

For the time being we confine ourselves to the study of random phenomena which are described by probabilistic models. A significant feature of these models is that they attempt to capture inherent uncertainty. How does one measure "uncertainty"? It may be pertinent to point out that the word information as used in daily parlance has two aspects "surprise" and "meaning". The mathematical theory of information is concerned with the surprise aspect and not with the semantic aspect of communication [5]. Accordingly, the quantity of information is related to the surprise element of information.

The information theory was developed in the context of the theory of communication to answer two fundamental questions :

(i) What is the ultimate achievable data compression?
(ii) What is the ultimate achievable rate of transmission of information?

For answering these questions, Claude Shannon in 1948 laid the foundation of information theory through his seminal paper "A mathematical theory of communication" [6]. He developed a measure of uncertainty which is referred to as entropy. However, early attempts in this direction were made notably by Nyquist in 1924 and Hartley in 1928 [7]. Though Shannon introduced entropy as a measure for problems in the context of communication theory, the measure has continued to find wide applications in problems from a large number of disciplines ranging over mathematical, physical, engineering, biological and social sciences. The reason for the remarkable success of entropy is due to the fact that one is able to quantify uncertainty present in probabilistic systems.

Gallager [8] notes : " Communication engineers adopted the idea and created the technology that led to today's information age. All communication lines are measured in bits per second, reflecting what Shannon had dubbed 'channel capacity' ". Shannon proved that it is possible to transmit information through a noisy channel at any rate less than channel capacity with an arbitrary small probability of error. Within information theory framework one seeks to achieve these ultimate limits of communication.

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In mathematical theory of communication, one is concerned with measure of transmitted information content from a known pool of possible messages. Hartley [7,9] described a logarithmic measure for information content for equally likely messages. What happens when messages are not equally likely? To answer this question, Shannon gave a measure of uncertainty known as Shannon's entropy, having the same mathematical expression as entropy in statistical mechanics.

1.3 Shannon's Entropy

Given an alphabet of n symbols $\{x_1, x_2, \ldots, x_n\}$, a random sequence $\{x_{i_1}, x_{i_2}, \ldots\}$ of independent observations therefrom can be regarded as an output of a discrete information source. Such a source produces messages by emitting a sequence of symbols from the source alphabet. Regarding the values in a sequence as successive observations on the variable X of a random experiment, we assume that the probability mass associated with the value x_i is p_i, i.e.,

$$p_X(i) \equiv P(X = x_i) = p_i, \quad i = 1, 2, \ldots, n.$$

The set $(X, \mathbf{p}) \equiv \{(x_1, p_1), (x_2, p_2), \ldots, (x_n, p_n)\}$, $\sum p_i = 1$ is called the source ensemble.The information associated with the outcome $\{X = x_i\}$ is denoted by

$$h(p_i) = -\log p_i. \tag{1.1}$$

It may be noted that the knowledge of the actual outcome of a random experiment removes the uncertainty attached to it and an equal amount of information is provided.

In (1.1), when logarithm base is 2, the information measured is in bits. With base e, for the natural logarithm the information is said to be measured in nats. Heuristic arguments based on probabilistic considerations can be advanced to establish the logarithmic nature of the measure of information. One may also derive this measure axiomatically from a set of natural postulates. Such a set of postulates will be discussed subsequently.

Definition The entropy $H(X) \equiv H_n(\boldsymbol{p})$ of the ensemble $\{(x_1, p_1), (x_2, p_2), \ldots, (x_n, p_n)\}$ as defined by Shannon is given by the expression

$$\begin{aligned} H(X) \equiv H_n(\boldsymbol{p}) &= H_n(p_1, p_2, \ldots, p_n) \\ &= -c \sum_{i=1}^{n} p_i \log p_i \,, \end{aligned} \tag{1.2}$$

where c is an arbitrary positive constant, generally taken as unity when logarithm base is 2 and entropy is measured in bits. To provide for impossible

events $0 \log 0$ is taken as zero. It is important to note that entropy

$$H_n(\mathbf{p}) = \sum_{i=1}^{n} p_i \; h(p_i) \tag{1.3}$$

is the average information. As X is a random variable with probability mass function p_X, (1.2) can be expressed as

$$H(X) = E[-\log p_X], \tag{1.4}$$

where E denotes expectation.

The expression $H(X)$ is related to entropy in thermodynamics. Recognizing the analogy, Shannon retained the same symbol as was used by Boltzmann in connection with his H–theorem [10]. In early nineteenth century entropy was introduced in the context of efficiency of heat engines. The basic quest in modern sciences has been to provide a microscopic basis for the macroscopic laws. The well-known formula $S = k \log \Omega$ relates the entropy S of a physical system to the number of microstates Ω accessible to the system. Here k is Boltzmann's constant. The microscopic entropy $S = -k \sum_{i=1}^{\Omega} p_i \log p_i$ yields $S = k \log \Omega$ if all microstates are equiprobable, i.e., $p_i = 1/\Omega$. The equation $S = k \log \Omega$ was so important and fundamental to Boltzmann that it was inscribed on his grave [11]. Chakrabarti and De [12] have considered the axiomatic characterization of Boltzmann and Gibbs entropies.

$H(X)$ can be interpreted as the average information provided by a realization of the experiment. Alternatively, one can say $H(X)$ is average uncertainty removed or average amount of information gained by observing the outcomes of X. For $n = 2$, we get Shannon's entropy function

$$f(p) \equiv H_2(p, 1-p) = -p \log p - (1-p) \log(1-p). \tag{1.5}$$

This can be interpreted as entropy of a Bernoulli variate X with parameter p and $H_2(p, 1-p)$ attains for $p = \frac{1}{2}$ its maximum value. Thus

$$H_2\left(\frac{1}{2}, \frac{1}{2}\right) = 1 \text{ bit.}$$

This is known as the normalization condition. It is worth observing that Shannon's entropy function is of basic importance as $H_n(\mathbf{p})$ can be expressed through it. Following notations as given in Mathai and Rathie [13],

$$H_n(\mathbf{p}) = \sum_{i=2}^{n} r_i \; f(p_i/r_i), \tag{1.6}$$

where

$$r_i = p_1 + p_2 + \ldots : + p_i, i = 1, 2, \ldots, n \quad \text{and} \quad \sum_{i=1}^{n} p_i = 1.$$

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where

$$r_i = p_1 + p_2 + \ldots + p_i, i = 1, 2, \ldots, n \quad \text{and} \quad \sum_{i=1}^{n} p_i = 1.$$

The function f in (1.5) satisfies the functional equation:

$$f(x) + (1-x) f\left(\frac{y}{1-x}\right) = f(y) + (1-y) f\left(\frac{x}{1-y}\right). \tag{1.7}$$

Here $x, y \in [0, 1]$ with $x + y \in [0, 1]$. The boundary conditions are $f(0) = f(1)$ and $f(\frac{1}{2}) = 1$.

1.4 Properties of Shannon's Entropy

Shannon's entropy $H_n(\mathbf{p})$ possesses several properties, viz., non-negativity, expansibility, symmetry, recursivity or branching principle, additivity, strong additivity, monotonicity. We give below mathematical expressions for some desirable properties.

(i) $H_n(\mathbf{p}) \geq 0$, with equality when $p(x_k) = 1$ and $p(x_i) = 0$ for all i, except $i \neq k$. Zero entropy implies the process is deterministic.

(ii) The entropy $H_n(p_1, p_2, \ldots, p_n)$ does not depend on the order of p_i's and under an arbitrary permutation $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ of the set $\{1, 2, \ldots, n\}$

$$H_n(p_1, p_2, \ldots, p_n) = H_n(p_{\alpha_1}, p_{\alpha_2}, \ldots, p_{\alpha_n}).$$

This property is known as symmetry.

(iii) $H_n\left(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n}\right)$ is a monotonic increasing function of n.

(iv) $H_n(\mathbf{p}) \leq H_n\left(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n}\right)$, equality holding only when $p_i = \frac{1}{n}$ for all $i = 1, 2, \ldots, n$. This implies that maximum entropy is attained when all probabilities are equal.

(v) H_n is a concave function of p_i's, its local maximum corresponds to global maximum as well.

(vi)

$$\begin{aligned} H_n(p_1, p_2, \ldots, p_n) &= H_{n-1}(p_1 + p_2, p_3, \ldots, p_n) \\ &\quad + (p_1 + p_2) H_2\left(\frac{p_1}{p_1 + p_2}, \frac{p_2}{p_1 + p_2}\right), \\ &\quad p_1 + p_2 > 0, \; n \geq 3. \end{aligned}$$

This property is known as recursivity.

(vii) Consider two probability distributions $P = (p_1, p_2, \ldots, p_n)$ and $Q = (q_1, q_2, \ldots, q_m)$ associated with independent random variables X and Y. The joint probability distribution of X and Y is

$$P(X = x_i, Y = y_j) = p_i q_j \;, \quad i = 1, 2, \ldots, n; \; j = 1, 2, \ldots, m.$$

Entropy of the joint distribution equals the sum of respective entropies associated with the independent random variables, i.e.

$$\begin{aligned} H_{mn}(p_1 q_1, \ldots, p_1 q_m; \ldots; p_n q_1 \ldots, p_n q_m) = \; & H_n(p_1, p_2, \ldots, p_n) \\ & + H_m(q_1, q_2, \ldots, q_m). \end{aligned}$$

This property of entropy is called additivity.

events $0 \log 0$ is taken as zero. It is important to note that entropy

$$H_n(\mathbf{p}) = \sum_{i=1}^{n} p_i \; h(p_i) \tag{1.3}$$

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where

$$r_i = p_1 + p_2 + \ldots + p_i, i = 1, 2, \ldots, n \quad \text{and} \quad \sum_{i=1}^{n} p_i = 1.$$

For other properties like strong additivity and sub-additivity, one may refer to Mathai and Rathie [13] and Kapur [14].

Note that, several attempts starting from Shannon have been made to uniquely characterize measure of uncertainty, making different sets of postulates. Notable attempts are due to Khinchin [15] and Fadeev [16]. An early attempt by Shannon subsumes the natural postulates. However the symmetry postulate (ii) has to be included to get the characterization theorem [13].

For reasonable requirements on the properties of a measure of uncertainty, it is worthwhile to note that it should be uniquely determined from four properties taken as axioms, viz monotonicity, additivity, recursivity and continuity of $H_2(p, 1-p)$ defined in (1.5).

At this point it is worth mentioning the connection of Shannon's entropy with Hartley entropy [9]. The property $H_n\left(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n}\right) = \log_2 n$ suggests that when every alternative of an n-state system has equal probability, then the Shannon entropy of the system is $\log_2 n$. This is nothing but the Hartley measure of information $H_H(n)$, which quantifies the amount of uncertainty that we face before we make a selection from a set of n alternatives. This uncertainty is resolved when the selection is made. Like Shannon's entropy, Hartley entropy can also be axiomatically derived based on the following three axioms :

Additivity : $H_H(n.m) = H_H(n) + H_H(m)$
Monotonicity : $H_H(n) \leq H_H(n+1)$
Normalization : $H_H(2) = 1$.

1.5 Asymptotic Equipartition Property (AEP)

Consider a discrete information source whose output block $\boldsymbol{x} = (x_{j_1}, x_{j_2}, \ldots, x_{j_N})$, x_{j_i} being the ith output, is a random sequence of N independent symbols (not necessarily different) from amongst a finite alphabet of n symbols. $\boldsymbol{x}$ may thus be regarded as a random sample of N independent outputs. Suppose that in the block $\boldsymbol{x}$, symbol x_j, $1 \leq j \leq n$, appears altogether r_j times. The vector $(r_1, r_2, \ldots, r_n)$ giving the number of occurrences is called the composition of the source word. If the symbol x_i of the alphabet occurs with probability p_i, then the probability of observing $\boldsymbol{x}$ is

$$p(\boldsymbol{x}) = \prod_{j=1}^{n} p_j^{r_j}, \quad \sum r_j = N.$$

By the strong law of large numbers $r_j/N \to p_j$ as $N \to \infty$, and we find that

$$\begin{aligned} -\frac{1}{N} \log\ p(\boldsymbol{x}) &\longrightarrow -\sum_{j=1}^{n} p_j \log p_j \quad \text{in probability} \\ &= \quad H(X) \end{aligned} \tag{1.8}$$

This is the well-known asymptotic equipartition property, which is a consequence of the weak law of large numbers. From (1.8) one finds

$$p(\boldsymbol{x}) \approx 2^{-NH(X)}.$$

The probability of observing a long string of elements becomes independent of the string and almost all large output sequences are equiprobable. AEP has implications in data compression where one attempts to find short descriptions of sequences of random variables [17,18].

1.6 Joint and Conditional Entropies, Mutual Information

We now consider a probabilistic system which can be described in terms of two random variables X and Y defined on the same probability space [5,7,19]. The joint entropy of X and Y is

$$H(X,Y) = -\sum_{i=1}^{n}\sum_{j=1}^{m} p_{ij}\ \log\ p_{ij}, \tag{1.9}$$

where $p_{ij} = p(x_i, y_j) = P(X = x_i, Y = y_j)$, $i = 1, 2, \ldots, n$ and $j = 1, 2, \ldots, m$ is the joint probability mass function. For convenience we write $P(X = x_i) \equiv p(x_i) \equiv p_i$ and $P(Y = y_j) \equiv q(y_j) \equiv q_j$. When X and Y are statistically independent, $p_{ij} = p_i q_j$, and it is easily seen that

$$H(X,Y) = H(X) + H(Y).$$

From general considerations, statistical dependence of random variables X and Y results in reduction of their joint entropy. Such a situation makes it relevant to introduce conditional entropy $H(X|y_j)$ representing the amount of uncertainty about X when outcome y_j is observed; symbolically

$$H(X|y_j) = -\sum_{i=1}^{n} p(x_i|y_j)\ \log\ p(x_i|y_j). \tag{1.10}$$

The conditional entropy $H(X|Y)$ is defined as the amount of uncertainty remaining about X given foreknowledge of Y. It is obtained by averaging (1.10) over all y_j's [7]. Thus

$$\begin{aligned} H(X|Y) &\equiv -\sum_{i=1}^{n}\sum_{j=1}^{m} q(y_j)p(x_i|y_j)\ \log\ p(x_i|y_j) \\ &= -\sum_{i=1}^{n}\sum_{j=1}^{m} p(x_i, y_j)\ \log\ p(x_i|y_j). \end{aligned} \tag{1.11}$$

For other properties like strong additivity and sub-additivity, one may refer to Mathai and Rathie [13] and Kapur [14].

Note that, several attempts starting from Shannon have been made to uniquely characterize measure of uncertainty, making different sets of postulates. Notable attempts are due to Khinchin [15] and Fadeev [16]. An early attempt by Shannon subsumes the natural postulates. However the symmetry postulate (ii) has to be included to get the characterization theorem [13].

For reasonable requirements on the properties of a measure of uncertainty, it is worthwhile to note that it should be uniquely determined from four properties taken as axioms, viz monotonicity, additivity, recursivity and continuity of $H_2(p, 1-p)$ defined in (1.5).

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$$p(\boldsymbol{x}) = \prod_{j=1}^{n} p_j^{r_j}, \quad \sum r_j = N.$$

By the strong law of large numbers $r_j/N \to p_j$ as $N \to \infty$, and we find that

$$\begin{aligned} -\frac{1}{N} \log\ p(\boldsymbol{x}) &\longrightarrow -\sum_{j=1}^{n} p_j \log p_j \quad \text{in probability} \\ &= \quad H(X) \end{aligned} \tag{1.8}$$

Similarly, we define

$$H(Y|X) = -\sum_{i=1}^{n}\sum_{j=1}^{m} p(x_i, y_j) \log q(y_j|x_i). \tag{1.12}$$

It is easy to see that

$$H(X|Y) = H(X,Y) - H(Y), \tag{1.13}$$

and

$$H(X|Y) \leq H(X), \tag{1.14}$$

with equality when X and Y are statistically independent.

For a better appreciation of the forgoing concepts, we consider X and Y as input and output respectively of a stochastic system while $H(X)$ represents the uncertainty about input X before output Y is observed, $H(X|Y)$ represents that after Y is observed. The difference can be regarded as a measure of dependence of X and Y and

$$I(X;Y) = H(X) - H(X|Y) \tag{1.15}$$

is called mutual information between X and Y [7].

From equations (1.8), (1.11) and (1.15), we easily deduce that

$$I(X;Y) = \sum_{i=1}^{n}\sum_{j=1}^{m} p(x_i, y_j) \log \left[\frac{p(x_i, y_j)}{p(x_i)q(y_j)}\right], \tag{1.16}$$

having properties

(i) $I(X;Y) = I(Y;X)$
(ii) $I(X;Y) \geq 0$,

with equality when X and Y are statistically independent. From Eqns. (1.13) and (1.15), one finds an interesting relation

$$I(X;Y) = H(X) + H(Y) - H(X,Y). \tag{1.17}$$

It may be mentioned that the capacity C of a continuous channel [7], when characterized by probabilistic dependence of output Y on input X, is given by

$$C = \max_{p(x)} I(X;Y).$$

where $p(x)$ represents all possible input distributions over which the maximum is taken.

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is called mutual information between X and Y [7].

From equations (1.8), (1.11) and (1.15), we easily deduce that

$$I(X;Y) = \sum_{i=1}^{n}\sum_{j=1}^{m} p(x_i, y_j) \log \left[\frac{p(x_i, y_j)}{p(x_i)q(y_j)}\right], \tag{1.16}$$

having properties

(i) $I(X;Y) = I(Y;X)$
(ii) $I(X;Y) \geq 0$,

with equality when X and Y are statistically independent. From Eqns. (1.13) and (1.15), one finds an interesting relation

$$I(X;Y) = H(X) + H(Y) - H(X,Y). \tag{1.17}$$

It may be mentioned that the capacity C of a continuous channel [7], when characterized by probabilistic dependence of output Y on input X, is given by

$$C = \max_{p(x)} I(X;Y).$$

where $p(x)$ represents all possible input distributions over which the maximum is taken.

1.7 Kullback-Leibler (KL) Directed Divergence

KL divergence provides a measure of distance between probability mass functions $P \equiv p_X(x)$ and $Q \equiv q_X(x)$ which are obtained under different operating conditions [20]. The distance of $p_X(x)$ from reference $q_X(x)$ is defined by

$$D_{KL}(P:Q) = \sum_x p_X(x) \log \frac{p_X(x)}{q_X(x)} \tag{1.18}$$

This distance is not symmetric in P and Q, i.e., $D_{KL}(P:Q) \neq D_{KL}(Q:P)$. One can show that $D_{KL}(P:Q) \geq 0$, with equality if and only if $P \equiv Q$. It is worth noting that KL divergence is related to mutual information which can be viewed as KL divergence between joint probability mass function $p_{XY}(x,y)$ and $p_X(x)q_Y(y)$.

1.8 Entropy of Continuous Distribution: Boltzmann Entropy

The entropy of a continuous random variable X with probability density function (pdf) $f(x)$ is defined as

$$h(X) \equiv h_B(X) \equiv h_B(f) = -\int_R f(x) \log f(x) dx \tag{1.19}$$

(under the assumption that the right hand side integral in (1.19) exists). R is the support set where $f(x) > 0$.

This entropy is named after Ludwig Boltzmann(1868) who used a similar expression for defining H-function in connection with the theory of gases [10]. The entropy $h(X)$ is referred to as differential entropy and in many ways it is similar to entropy of a discrete random variable. Though $h(X)$ has a form analogous to that for a discrete random variable, it needs to be emphasized that (1.19) does not provide the limiting form of the discrete entropy. To illustrate this aspect we follow the treatment in Ihara [21] and consider an approximating discrete random variable X_n defined by

$$X_n = \frac{k}{2^n}, \qquad \frac{k}{2^n} \leq X < \frac{k+1}{2^n}, \qquad k = 0, \pm 1, \pm 2, \ldots .$$

Setting

$$\Delta_{n,k} = \{x;\ \frac{k}{2^n} \leq X < \frac{k+1}{2^n}\}$$

and

$$\rho_{n,k} = \int_{\Delta_{n,k}} f(x) dx. \tag{1.20}$$

The entropy of X_n is given by

$$H(X_n) = -\sum_{k=-\infty}^{\infty} \rho_{n,k} \log \rho_{n,k}. \tag{1.21}$$

As proved in Ihara [21], under appropriate assumptions on the pdf $f(x)$,

$$\lim_{n\to\infty} \{H(X_n) - n\log 2)\} = h(X), \tag{1.22}$$

while $H(X_n)$ diverges as $n \to \infty$.
A basic question relates to a reconciliation of discrepancy between Shannon's and Boltzmann's entropies. One can get rid of divergence term in (1.22) if one considers the difference of entropies of two probability distributions. It is shown in Ihara [21] that

$$\lim_{n\to\infty} \{H(X_n) - H(Y_n)\} = h(X) - h(Y), \tag{1.23}$$

where Y is a real random variable satisfying the same assumptions as those for random variable X. Following Klir and Folger [22], it is easy to check that the Boltzmann information transmission

$$\begin{aligned} T_B(f(x,y)|x \in [a,b], y \in [\alpha,\beta]) = &-\int_a^b f_X(x)\log_2 f_X(x)dx \\ &-\int_\alpha^\beta f_Y(y)\log_2 f_Y(y)dy \\ &+\int_a^b\int_\alpha^\beta f(x,y)\log_2 f(x,y)dxdy \end{aligned} \tag{1.24}$$

yields a legitimate limiting form for $I(X,Y)$ as given in (1.17).

Similarly, Kullback-Leibler measure of directed divergence has a genuine Boltzmann counterpart given by

$$B(f(x), f'(x)|x \in [a,b]) = \int_a^b f(x)\log\frac{f(x)}{f'(x)}dx, \tag{1.25}$$

where $f(x)$ and $f'(x)$ are two density functions defined on $[a,b]$.

1.8.1 Boltzmann H-theorem

For the sake of completeness, we now briefly mention the H-theorem which provided a molecular basis for the natural tendency of a physical system to approach equilibrium . In 1872 Boltzmann introduced the H-function

$$H(t) = \int_{-\infty}^{\infty} f(\boldsymbol{v},t)\log f(\boldsymbol{v},t)d\boldsymbol{v}, \tag{1.26}$$

The entropy of X_n is given by

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where $f(\boldsymbol{v},t)$ is velocity distribution function of gas molecules [21]. Boltzmann used his transport equation to prove that $\frac{dH}{dt} \le 0$, implying that the H-function decreases with time or stays constant. Moreover $\frac{dH}{dt} = 0$ defines the equilibrium, when the gas molecules have Maxwell-Boltzmann distribution, which according to Boltzmann, is a necessary and sufficient condition for equilibrium.

Boltzmann's H-theorem met with a strong opposition at that time because it seemed to contradict the recurrence theorem due to Poincaré. This violent criticism of H-theorem led Boltzmann to a state of despair which gets reflected in Boltzmann's book "*Vorlesungen über Gastheorie*". He wrote:

> *"It will be a great tragedy of science if the theory of gases were temporarily thrown into oblivion because of momentary hostile attitude towards it... I am conscious of being only an individual struggling weakly against the current of time. But it still remains in my power to make a contribution in such a way that, when the theory of gases is again revived, not too much will have to be rediscovered".*

Ihara [21] mentions that Boltzmann's growing pessimism led him to commit suicide in 1906 – one of the greatest tragedies in the history of physics. Ihara further states that soon after Boltzmann's death, kinetic theory and his statistical concepts were accepted. As early as 1872, Boltzmann in his work on the kinetic theory of gases, laid the foundation of probabilistic theory of entropy.

1.9 Entropy and Applications

1.9.1 Applications to logical problems

The concepts of entropy have been successfully employed to obtain solutions of certain type of logical problems when all information can be expressed on the basis of yes-no answers [23,24]. In his 'A diary on information theory', Rényi [24] poses several problems based on 'Bar Kochba' game for introducing the mathematical notion of information. This game requires all information in yes-no answers which can be coded as a of sequence of two symbols. For the purpose of illustration we consider below some examples.

(i) **Identification of a particular student:** Suppose that in a class of 32 students, the teacher has a certain student in his mind. At least how many apt questions with yes-no answers can be asked of the teacher to find out who that student is? We know that amount of information needed to specify one amongst 32 possibilities is $-\log_2(1/32)$, i.e., 5 bits. A question which can be answered 'yes-no' contains $-\log_2(1/2)$, i.e., 1 bit of information. Accordingly 5 apt questions can serve the purpose. Formulating and solving Bar-Kochba type of problems, Rényi develops from basic arguments Hartley's formula [24]. Suppose that about an unknown element

we know nothing except that it belongs to a set having N elements. This information amounts to $\log_2 N$ bits.

(ii) **Counterfeit coin problem**: This is a well-known problem (Blake [25]) which requires to identify one counterfeit (defective) coin from a collection of 9 (otherwise) identical coins. What is the minimum number of weighings required? We follow the solution given in [25]. As any of the nine can be defective (lighter or heavier), there are 18 possibilities. To specify the defective coin, one needs $\log_2 18 = 4.170$ bits. In a weighing there are three possibilities, either the balance tips left or right or balances. The maximum information is obtained when three possibilities are equiprobable and then it is $\log_2 3 = 1.585$ bits. Thus it is quite possible to identify the defective coin in three apt weighings.

It may be noted that the additional information about the defective coin being heavier admits only of nine possibilities instead of eighteen. Consequently, to identify the defective one we need $\log_2 9 = 3.170$ bits. Thus it is possible to find the defective one in two weighings.

(iii) **Sorting problem** : The sorting problem deals with arranging in increasing (or decreasing) order a set of n distinct elements $x_1, x_2, \ldots, x_n$. An important question is to obtain an estimate of the minimum number of comparisons needed to sort them. There being $n!$ ways of arranging n elements, the information needed to specify the sorted arrangement is $\log n!$ bits since the maximum information obtained from each comparison is $\log_2 2 = 1$ bit. Thus for large n, the number of comparisons needed $\approx \log n!$.

Using Sterling's approximation $n! \sim \sqrt{2\pi} \quad n^{n+(1/2)}\, e^{-n}$, we have

$$\log_2 n! = O(n \log\ n). \tag{1.27}$$

Accordingly, it is not impossible to sort n distinct elements in $O(n \log n)$ comparisons, a remarkable result which could be so easily obtained.

1.9.2 Data compression and coding

Entropy plays a pivotal role in the design of algorithms for data compression and provides a fundamental lower bound to compression. In data compression, often referred to as coding, one requires to eliminate significant redundancy present in signals. Coding techniques are needed for transmission or storage of data. Data compression can be classified into two categories, viz lossless compression and lossy compression. The objective of the former is to represent the signal in terms of minimum number of bits such that one is able to recover the original signal from the decompressed data while in the case of lossy compression, the goal is to achieve the best possible fidelity in the decompressed data approximating uncompressed data allowing a prescribed amount of information loss.

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Defining the prefix code, which is instantaneously and uniquely decodeable, we wish to construct a code with minimum average length. Let random variable X assume values $x_1, x_2, \ldots, x_n$ with respective probabilities $p_1, p_2, \ldots, p_n$. Consider a binary code (a sequence of 0's and 1's) for each x_i having code length l_i, then the expected length of the code word is

$$E[L] = \sum_{i=1}^{n} p_i l_i.$$

The set of all the possible code words satisfies Kraft's inequality [17]:

$$\sum_{i=1}^{n} 2^{-l_i} \leq 1,$$

which is a necessary and sufficient requirement for the existence of prefix code. Conversely, if code lengths l_i's satisfy this inequality, then one can construct a prefix code with these lengths. A significant result giving bounds on optimal code length is

$$H(X) \leq E[L] \leq H(X) + 1. \tag{1.28}$$

For a lossless compression, the lower bound is given by entropy which determines the lower bound on the average length of the shortest description of a given random variable.

Kolomogorov's Complexity and Shannon's entropy
In passing it may be mentioned that Kolomogorov introduced the notion of algorithmic (descriptive) complexity as the minimal description length for a non-random object. Thus, the objective is to find the shortest description of a binary computer program for computing the string [17]. Kolomogorov complexity does not require probability distribution for its definition, and is more general than Shannon's entropy which gives the lower bound on average length of the shortest description of a random variable. An interesting fact as discussed by Cover and Thomas [17] reads: "The expected length of the shortest binary computer description of a random variable is approximately equal to its entropy. Thus the shortest computer description acts as a universal code which is uniformly good for all probability distributions".

1.9.3 Entropy of language and redundancy

Entropy of English language
Suppose we are given a discrete memoryless information source which generates statistically independent symbols from the source alphabet $X = \{x_1, x_2, \ldots, x_n\}$. The source is assumed to be stochastic such that each symbol occurs with certain probability, i.e., $p_i = P(X = x_i)$. A more realistic situation would require to deal with the dependence of symbols generated from information source having memory.

A group of consecutive symbols comprises a word. For example in English language the source alphabet consists of 26 letters (symbols) and a space symbol. The text can be regarded as information source. If we consider language as a stochastic process, how well does the source in English relate to stationary stochastic process? Related to this is the question of various stochastic approximations which can generate texts in English in an increasing order of resemblance.

The pioneering effort for probabilistic analysis of language was made as early as 1913 by Markov. He analyzed Pushkin's Eugene Onegin to study whether an upcoming letter (symbol) is consonant or vowel through a mathematical model known as Markov chain. Extensive studies have been carried out to compute the probability of a complete string of words say $w_1, w_2, \ldots, w_n$. In bi-gram one employs Markov assumption to approximate the conditional probability $P(w_n|w_1, w_2, \ldots, w_{n-1})$ of occurrence of word w_n given all $n-1$ words by conditional probability $P(w_n|w_{n-1})$ of occurrence of word w_n given only the preceding word. One can extend this framework from bigram to trigram where one considers dependence on two preceding words instead of one and the resulting model is a second-order Markov model. In a similar manner N-gram models can be constructed which have been successfully employed by IBM speech recognition laboratory [26]. These ideas have led to construction of more sophisticated N-gram, such as variable length N-grams in language modelling and speech recognition [27]. Various stochastic approximations have been used to evaluate N-gram systems through entropy based metrics. A pertinent question is to ask relevance of entropy of English language. Jurafsky and Martin [27] argue : "One reason is that the true entropy of English would give us a solid lower bound for all of our future experiments on probabilistic grammars. Another is that we can use entropy values for English to help understand what parts of a language provide the most information (for example, is the predictability of English mainly based on word order, on semantics, on morphology, on constituency or on pragmatic cues?). This can help us immensely in knowing where to focus our language modelling efforts".

For the purpose of illustration we consider English language. In fact the most common letter is E having a relative frequency of 0.13 and the least occurring are Q and Z with a relative frequency of 0.001. The letter E occurs so frequently that it is highly unlikely to find a sentence not containing it. We also do not witness uniformity in the frequency of pair of letters either. It is commonly observed that the letter Q is invariably followed by U. Of course the most frequent pair encountered is TH which has a relative frequency of about 0.037. This fact can be best illustrated by examples of Markov approximations to English language considered by Shannon [6]. In the zeroth order approximation, where the symbols are considered to be independent and equiprobable, the entropy turns out to be $\log_2 27 = 4.78$ bits. As we increase the order of approximation, the entropy starts decreasing. For example

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in the fourth order approximation corresponding to frequency of quadruplets of letters matching English text, the entropy reduces to 2.8 bit/letter. Increasing the order of Markov approximation, the resemblance with English text gets clearer [17].

Shannon [28] also made an extensive study on the predictability of the English language, i.e., how well one can predict the next letter given the preceding n letters. Shannon estimated both the upper and lower bounds for entropy and redundancy of English language. Shannon's analysis suggested that in ordinary literary English, the long range statistical effects up to 100 letters reduce the entropy to about 1 bit per letter with a corresponding redundancy of 75%. The redundancy could still be higher when structure extending over paragraphs, chapters etc. is considered.

Redundancy

The development of stochastic models for language can be used for text compression in view of the fact that languages have inherent redundancy. The redundancy of a discrete memoryless source [7] is defined as:

$$R = 1 - \frac{H(X)}{max\{H(X)\}} = 1 - \frac{H(X)}{\log_2 n} \tag{1.29}$$

where $H(X)$ is the entropy of the source and $max\{H(X)\} = \log_2 n$ when all symbols are equiprobable with $p_i = \frac{1}{n}$ for all $i = 1, 2, \ldots, n$. R is a dimensionless number lying between 0 and 1.

When there is dependence among symbols, the information content is reduced. The redundancy for source with memory is obtained in terms of $H_\infty(X)$. To define $H_\infty(X)$, we consider messages instead of individual symbols. Defining information content $H(Y)$ of message Y comprising N symbols, i.e.,

$$H(Y) = H(X_1, X_2, \ldots, X_N) \quad \text{bits/message},$$

the amount of entropy rate per symbol is given by

$$H_N(X) = \frac{1}{N} H(Y) = \frac{1}{N} H(X_1, X_2, \ldots, X_N).$$

In the limit, $H_N(X)$ converges such that

$$H_\infty(X) = \lim_{N \to \infty} H_N(X),$$

denoting entropy rate (or per symbol entropy).The dependence redundancy is defined as

$$R_\infty = 1 - \frac{H_\infty(X)}{H(X)}.$$

Redundancy, in case of optimal coding, has special significance as it provides an estimate of the proportion of dispensable text [24]. Shannon's determination of this proportion in the English text as $R = 0.5$ implies that the text

can be reduced to half its size by omitting all possible redundancy therein. The question then arises about the desirability of redundancy in text. It enables one to understand the text by reducing the effect of possible mistakes which might have crept into the text. Rényi [24] has very clearly brought its significance and observes that we would not be able to talk to each other if languages were not strongly redundant. Further he notes that natural languages can be regarded as natural error correcting code.

1.9.4 Large deviation theorem

It is interesting to note that entropy, minimum relative entropy occur in large deviation principle (LDP), which describes the rate of convergence to zero probability. Large deviation property can be regarded as a kind of limit theorem in probability [21]. LDP is concerned with estimating properties of rare events and is finding useful applications in communication networks [29]. LDP allows us to analyse situations involving large deviations from typical situations. In any probabilistic system, large deviations from typical situations are of great concern. For example, in a communication network, a sudden burst of traffic is a typical situation and the large deviation theory enables us to analyse the performance of the network in such situations.

1.9.5 Cooperative Behaviour in Innovation Diffusion

Bartholomew [30] proposed a stochastic model of diffusion of innovation in a social group of size N. The innovation diffuses through a combination of twin processes, viz. mass mediated and interactive processes. Bartholomew took into consideration the possibility that individuals diffusing the innovation are liable to lose interest in the matter and assume the status of ignorants. Of course, at a subsequent time he/she may find his/her interest rekindled through the twin processes. It may be mentioned that the word innovation is quite comprehensive and it may mean news, opinion, rumour, 'information', new product etc. Defining $n(t)$ as the number of active spreaders at time t, the diffusion process $n(t)$ can be modelled as a birth-death process with transition probabilities

$$\begin{aligned} Pr(n \to n+1 \text{ in } (t, t+\delta t)) &= \{\alpha(N-n) + \beta n(N-n)\}\,\delta t + o(\delta t) \\ Pr(n \to n-1 \text{ in } (t, t+\delta t)) &= \mu_n \delta t + o(\delta t), \quad n = 1, \cdots, N \end{aligned}$$

where α and β denote transition intensities due to mass-mediated process and interactive process respectively while μ^{-1} represents the mean duration of time for which spreader remains active [31]. A more general class of stochastic models of information diffusion with quadratic nonlinearities is considered by Sharma *et al.* [32].